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Brief Interactive Virtual Reality Mindfulness Training with Real-Time Biofeedback for Anxiety Reduction: A Pilot Study

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Abstract
Recent advancements in virtual reality (VR)-enhanced mindfulness training have demonstrated efficacy in reducing anxiety, particularly among novice practitioners and university students. However, existing VR mindfulness interventions predominantly rely on static natural environments without incorporating interactive elements, limiting their capacity to engage users dynamically. To address this gap, this study introduces an innovative interactive VR mindfulness system, where real-time heart rate variability (HRV) biofeedback dynamically modulates the virtual environment (e.g., fog dissipation, auditory cues, and visual transformations), creating a responsive and immersive experience. We investigated the effects of a single five-minute interactive VR mindfulness session on anxiety, mindfulness states, and HRV, compared to traditional audio-based mindfulness and an non-intervention control. Seventy-five university students with recent anxiety symptoms were randomized into three groups: interactive VR mindfulness (VR-MG), traditional mindfulness (MG), and control (CG). Prior to and following the five-minute intervention, anxiety, mindfulness levels, and physiological data (HRV) were measured for all three groups, with repeated measures analysis of variance performed. The results showed that the five-minute VR mindfulness interactive training significantly improved participants' anxiety levels, enhanced their mindfulness states, and effectively increased HRV, whereas traditional mindfulness and control conditions showed no significant HRV changes. Overall, this study indicates that brief VR mindfulness interactive training has significant effects on improving anxiety and mindfulness states, as well as enhancing HRV, highlighting its potential for psychological health interventions.

Keywords Mindfulness · Virtual reality · Biofeedback · Brief mindfulness intervention · Heart rate variability · Anxiety

Introduction

Mental health challenges among university students have emerged as a growing global concern (Tabor et al., 2021). A meta-analysis involving 100,187 participants found that the global pooled prevalence rate of anxiety symptoms among university students was 39.0% (Li et al., 2022). University life, widely regarded as a pivotal developmental stage, exposes students to multifaceted stressors, including academic demands, interpersonal conflicts, and career uncertainty, which collectively contribute to heightened academic, social, and employment-related anxiety (Zhu, 2023). Notably, adolescence and early adulthood (ages 12–24) are recognized as critical windows for the onset of mental health disorders, with approximately 75% of lifetime diagnoses emerging before age 25 (Li et al., 2022). Chronic anxiety not only impairs social engagement and skill development (Wu et al., 2013), but also undermines occupational performance (Ford et al., 2011), and increases susceptibility to chronic illnesses (Orme-Johnson & Barnes, 2014). These cascading effects underscore the profound societal repercussions of anxiety, impacting nations, educational institutions, families, and individuals (Liu et al., 2023). Consequently,

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addressing anxiety in university students demands urgent scholarly and policy attention.

Mindfulness is a concept originally derived from Buddhism as a practice of the Noble Eightfold Path, developed from practices such as seated meditation, contemplation, and insight, encompassing awareness, attention, and remembrance (Langer et al., 1978). Mindfulness is defined as moment-to-moment, non-judgmental awareness, developed by paying attention in a specific way—namely, in the present moment—with the highest possible openness, non-reactivity, and non-judgment (Kabat-Zinn, 2015). Kabat-Zinn introduced mindfulness into clinical practice and coined the term "Mindfulness-Based Stress Reduction (MBSR)." Initially applied to alleviate chronic pain, MBSR has since expanded to address negative emotions such as anxiety and depression. MBSR is typically implemented in group settings, lasting eight weeks, with sessions held once a week for approximately 2–2.5 h. Each group consists of around 30 participants, led by qualified professionals who also assign homework to encourage ongoing mindfulness practice and long-term adherence (Kabat-Zinn, 2003). Mindfulness practices have been shown to be effective in enhancing learning abilities, reducing stress, regulating emotions, and improving human well-being (Niedermeier et al., 2022; Nyklíček & Kuijpers, 2008; Vorontsova-Wenger et al., 2022; Yun et al., 2020). Nevertheless, mindfulness still faces several challenges and limitations in its practical application. Many studies have reported a generally low adherence rate to mindfulness interventions (Brooks et al., 2021; Lyzwinski et al., 2019), as well as a high dropout rate (Khoury et al., 2013; Nam & Toneatto, 2016). These issues are particularly prominent in the application of traditional mindfulness training, especially among groups such as university students, affecting its long-term efficacy and widespread implementation. Traditional mindfulness practices typically involve regular study groups and an eight-week course, which may not be suitable for many individuals, particularly students. With the heavy academic workload and life pressures students face, maintaining motivation to participate in mindfulness courses over the long term is challenging. Moreover, the time and financial demands of traditional courses limit their accessibility (Modrego-Alarcón et al., 2021). Additionally, one of the core elements of mindfulness practice is the continuous regulation of attention, which can be especially difficult for beginners or individuals with high emotional instability. Due to external distractions and personal emotional fluctuations, beginners often find mindfulness practice abstract and complex, making it hard to sustain (Lutz et al., 2008; Navarro-Haro et al., 2017). As a result, to improve mindfulness training's appeal and adherence among university students, researchers are exploring more flexible, accessible methods. By incorporating new technologies like virtual reality (VR), they aim to create immersive, interactive experiences that enhance student motivation, reduce dropout rates, and increase adherence, all while minimizing the time and financial burdens of traditional programs.

Over the past decade, mindfulness training has been increasingly supported by information and communication technology (ICT) solutions, with scientific literature addressing this trend. Among these technologies, virtual reality (VR), particularly immersive VR, has proven to be especially beneficial for enhancing mindfulness practices (Arpaia et al., 2022). VR refers to a computer-generated simulation of a real-world environment, where users experience an interactive environment through sensory stimuli via VR headsets, creating a sense of presence that allows them to feel as though they are physically immersed in a specific scene (Bryant et al., 2020). This immersive experience fosters mindfulness and concentration (Cipresso et al., 2018; Slater, 2018). The combination of virtual environments with guided mindfulness audio has been shown to enhance the sense of presence compared to simple narration, and VR's ability to offer personalized environments further increases engagement and motivation for practice (Lu et al., 2023). Moreover, VR provides a distraction-free space, making it easier for individuals to concentrate on their mindfulness practice (Costa et al., 2019). Therefore, VR-based mindfulness training can improve both participant motivation and adherence.

Currently, many clinical studies have shown that mindfulness-based VR interventions significantly outperform traditional mindfulness practices. For example, in individuals with chronic pain, a virtual reality system has been shown to be more effective in reducing reported pain levels compared to a control group that only listened to guided mindfulness audio tracks (Gromala et al., 2015). However, in non-clinical settings, the effects are comparable to traditional mindfulness (Tarrant et al., 2022): for instance, Modrego-Alarcón et al. (2021) randomly assigned 280 students to either "MBP," "MBP+VR," or a "Relaxation" (active control) group. The results indicated that both the "MBP+VR" and "MBP" groups effectively improved perceived stress and anxiety, but there were no significant differences between the two. We believe that the limited effectiveness may be due to the insufficient interactivity of VR systems, as most studies primarily rely on the use of natural environments without incorporating additional interactive elements.

In the context of virtual reality (VR) interactive technology, biofeedback has been recognized as a promising enhancement to VR-supported mindfulness interventions (Arpaia et al., 2022). Biofeedback is a non-invasive psychophysiological treatment technique that monitors users' physiological activities and provides real-time feedback

by extracting relevant parameters. Common biofeedback data include skin electrical activity, respiration rate, blood flow changes, and brain electrical activation (Riva et al., 2017). In immersive virtual reality, biofeedback not only enhances external sensory experiences but also reflects users' internal physiological states, fostering awareness and promoting a mindful state. Among the many biofeedback parameters, heart rate variability (HRV), as a measure of autonomic nervous system flexibility, has garnered our attention. Higher HRV is typically associated with stronger emotional regulation and better cognitive performance (Appelhans & Luecken, 2006; Hansen et al., 2003, 2009; Ottaviani et al., 2019), whereas low HRV is linked to various psychopathological conditions (Beauchaine & Thayer, 2015). Studies have shown that mindfulness practice can significantly improve HRV, with this change potentially arising from the dominance of the parasympathetic nervous system, particularly through respiration changes regulated by the vagus nerve (Grossman & Taylor, 2007; Shaffer et al., 2014), as respiration regulation is central to mindfulness practice (Kabat-Zinn, 2003). In fact, studies have demonstrated that during mindfulness, respiration rate decreases, and HRV response increases (Adler-Neal et al., 2020; Ahani et al., 2014). Furthermore, Failla et al. (2022) notes that the application of VR-enhanced mindfulness interventions in non-clinical populations is still in its early stages, with many studies lacking physiological measurements such as HRV. Therefore, we plan to utilize HRV as a physiological marker to observe mindfulness and integrate real-time biofeedback into the VR system to further enhance the effectiveness of mindfulness interventions.

In traditional mindfulness interventions for anxiety, the most commonly used program is MBSR (Mindfulness-Based Stress Reduction), which typically lasts for eight weeks, requiring participants to engage in daily practice and attend weekly group meetings (Kabat-Zinn, 2003). This long-term commitment and the weekly time investment can pose challenges for many individuals who could benefit from mindfulness, particularly for those in fast-paced, high-demand groups (Craigie et al., 2016; Crane & Kuyken, 2013).For groups such as busy college students or individuals with high work-related stress, traditional mindfulness interventions may not be easily adaptable. These groups often need rapid and easily implemented interventions to apply mindfulness techniques flexibly in daily life. As a result, brief mindfulness interventions (MBIs) have gained attention in recent years. A review by Howarth et al. (2019) of 85 brief MBIs revealed that brief mindfulness interventions (with an average duration of 15 min) showed significant variability in health outcomes, with some studies reporting non-significant health benefits (Howarth et al., 2019). Nevertheless, brief mindfulness interventions are still considered an important avenue for exploration, particularly in time-constrained situations.

To further explore the effectiveness of mindfulness interventions within a short time frame, we designed an innovative five-minute brief mindfulness VR intervention. This intervention leverages virtual reality technology to provide an immersive experience, while integrating real-time physiological feedback, such as heart rate variability (HRV), to offer participants immediate feedback on their physiological state. Research has shown that five-minute mindfulness practices can significantly reduce anxiety and improve emotional regulation and cognitive performance (Greeson, 2009; Jha et al., 2010). Through VR technology, we aim to further optimize the experience, providing an efficient, convenient, and easily implementable tool for mindfulness intervention.

To this end, we designed this study to assess the effects of a five-minute mindfulness VR training on mindfulness and anxiety in a sample of university students. Additionally, we evaluated the potential mediating role of HRV in the relationship between mindfulness and anxiety. We tested the following hypotheses: (1) A single five-minute VR mindfulness training session can effectively reduce anxiety, improve mindfulness levels, and enhance HRV; (2) A single five-minute VR mindfulness training session is more effective than traditional mindfulness; (3) HRV mediates the improvement in anxiety induced by mindfulness.

Materials and Methods

Participants and Design

Prior to data collection, utilizing G*Power3.1 to conduct a priori power analysis, the smallest number of participants needed to detect a medium-sized correlation of 0.25 with a power of 0.95 and an alpha of 0.05, two-tailed, was found to be sixty-six ($n = 22$ per group). Initially, we recruited participants through online promotions and offline posters aimed at promoting mental health. The inclusion criteria were as follows: (a) aged between 18 and 28 years; (b) experienced episodes of anxiety over the past week; (c) no history of diagnosed psychiatric disorders (e.g., major depression, anxiety disorders) or smoking addiction; and (d) provide written informed consent before participating in this study. Exclusion criteria included those: (a) who had undergone mindfulness training; and (b) who had anxiety levels during the experiment with a baseline STAI-S score < 43 (Dai et al., 2023). The subjects could complete an appointment time for offline experimental participation using the online contact form.

In the end, seventy-nine university students from our university were recruited as experimental subjects. Unfortunately, four participants were excluded from the study due to insufficient levels of anxiety during the experiment (STAI-S baseline score < 43). Seventy-five subjects participated in the full intervention and experiment, 28 males and 45 females. The mean age of the participants was 20 years.

The study employed a 2 × 3 mixed-design, with the independent variables being time (pre-test and post-test) and group, consisting of the virtual reality mindfulness group (VB-MG), the mindfulness group (MG), and the control group (CG). All participants provided informed consent prior to the study and received compensation for their participation. The study protocol was approved by the Institutional Review Board.

Procedures

This study was introduced to the prospective participants as a mental health promotion study, where subjects were not aware of their group allocation prior to the intervention to reduce self-selection bias (single blind). Participants were informed about the procedures for the attachment of the physiological instruments (Polar H10) the day before the experiment. They were instructed to refrain from alcohol and nicotine consumption prior to the heart rate variability (HRV) measurement to minimize the known effects of these factors on autonomic activity. Additionally, participants were advised to avoid engaging in intense physical activity to prevent any potential interference with the HRV data. After signing the informed consent form, anxiety levels in the participants were measured by inducing anxiety through a recall technique, while their heart rate variability data under anxiety were simultaneously recorded using the Polar H10 device. The recall technique involved asking participants to recall relevant personal memories, which would trigger anxiety for approximately five minutes. After inducing anxiety in the participants, they were required to complete the The state version of the State-Trait Anxiety Inventory (STAI-S) and the Toronto Mindfulness Scale (TMS). If the participant did not meet the requirement (STAI-S < 43), they were compensated and asked to exit the trial. After the pre-test, a random sample by drawing lots was used to allocate participants on a 1:1:1 basis to the virtual reality mindfulness group, the mindfulness group, and the control group, followed by a five-minute intervention. During the five-minute intervention, participants' HRV responses were also recorded. Upon completion of the five-minute intervention, participants were asked to complete the STAI-S and TMS once again. Participants in the VR-MG group were also required to complete the Igroup Presence Questionnaire (IPQ) and undergo a brief interview in which they expressed their feelings during the experience. The interview lasted approximately 5 min (Fig. 1).

Interventions

Virtual reality mindfulness intervention

The VR-MG was instructed to practice mindfulness within the interactive VR environment (Forest of the soul) designed for this study. The mindfulness intervention practice (https://www.headspace.com/) was adapted from E pisode 4 of the "Headspace Meditation and Mindfulness Guide," and included focusing on a chosen object (such as the body or breath), monitoring mental activity, noticing distractions, and cultivating a non-judgmental attitude towards one's experiences (i.e., calmness). The selected segment had a total duration of approximately 5 min and was a complete mindfulness practice session. The Headspace app has been applied in previous research demonstrating effects pertaining to stress-relief such as overt self-reported stress, self-reported well-being and self-reported mindfulness (Gavrilova & Zawadzki, 2023). All participants used the EDIFIER W800BT Plus wireless Bluetooth over-ear headphones for the audio experience to ensure an enhanced and optimal experience.

For the VR experience, we built the virtual environment using the Unity 3D development engine and loaded it into the HTC VIVE headset. Forest of the Soul was a mindfulness meditation scene designed for this study. Participants found themselves in a forest enveloped in dense fog, with a flower floating in the air, surrounded by a ring of light particles. These particles expanded and contracted at a steady frequency. A guided mindfulness audio gently prompted participants to focus on their breathing and relax their bodies. Faint sounds of flowing water and bird songs could be heard, hidden within the mist. As participants became increasingly relaxed and immersed, they noticed the fog gradually lifting. The scene became clearer, revealing a vast field of flowers and lush green mountains. The floating flower began to bloom slowly, and the sounds of flowing water and birds became more distinct, as if all the haze and negative emotions were fading away. Overall, the scene featured tranquil and reassuring imagery, such as towering trees in the distance, a river flowing gently, and flowers swaying in the breeze. The scenes and their elements are selected based on Attention Restoration Theory (ART) and Stress Recovery Theory (SRT). These theories suggest that immersion in natural environments can evoke positive emotions and restore attention resources depleted by stress (Kaplan, 1995). The viewpoint was free to rotate with the participant's head, with no limitations.

Fig. 1 Flowchart of participants

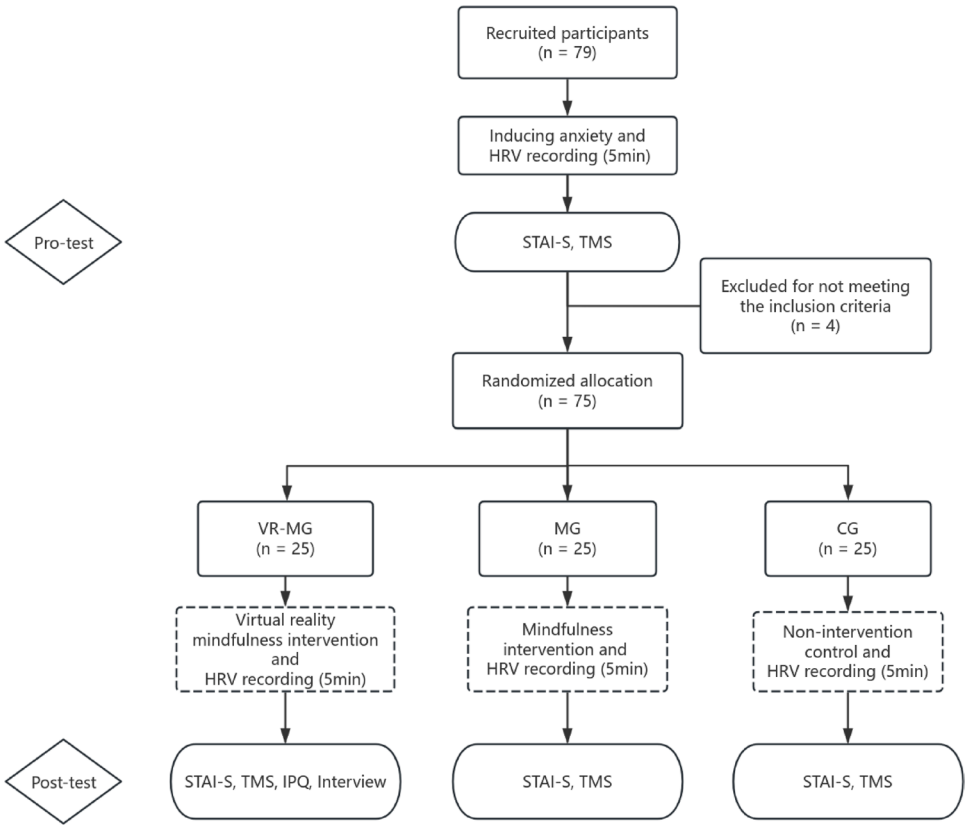

Fig. 2 Changes in VR scenes corresponding to fluctuations in heart rate variability

The interactive design during the experience was as follows: The light particles surrounding the floating flower in the air expanded and contracted every four seconds to guide the user into deep breathing. The biofeedback in the scene was reflected through the positive changes in the environment based on the participant's HRV physiological data. In the bottom right corner of the scene, there was a progress bar that provided real-time feedback on the HRV value. This data was monitored by an experienced instructor observing the HRV feedback from a chest strap worn by the participant. The instructor could adjust the HRV data by pressing the "K" key, which increased the progress bar by 5 units. As the HRV increased, the fog gradually dissipated, and the sounds of flowing water, birdsong, and soothing music became louder. Simultaneously, the floating flower slowly began to bloom. The surrounding scene started to reveal its true colors, and the view became clearer and broader (Fig. 2). Based on Kabat-Zinn's research, mindfulness was a practice of "non-attachment to the outcome." Therefore, there were no obvious punishments or negative feedback in the experience, nor were there any undesirable endings. Participants could always complete the mindfulness session smoothly. Even if the HRV level did not increase, in the final thirty seconds of the session, all the flowers bloomed, and the fog dissipated.

Mindfulness intervention

Participants in the MG group listened to the same mindfulness training audio track as those in the VR-MG group. The duration of the mindfulness exercise was approximately 5 min.

Non-intervention Control

Participants in the control group listened to a natural science audio track, which consisted of the first chapter of A Brief History of Time, also lasting 5 min.

Measures

The State Version of the State-Trait Anxiety (STAI-S)

The state version of the State-Trait Anxiety (STAI-S) was used to measure participants' state anxiety. The scale consists of 20 items (e.g., "I feel calm") and employs a four-point rating scale (1 = Not at all calm to 4 = Very calm). Items related to positive emotions are reverse-scored, and the scores for all items are summed. The total score represents the participant's level of anxiety, with higher scores indicating higher levels of anxiety (Spielberger et al., 1983).

Toronto Mindfulness Scale (TMS)

The Toronto Mindfulness Scale (TMS) was used to measure participants' state mindfulness. The scale includes 13 items and employs a four-point rating scale (0 = Not at all to 4 = Very much). The scale consists of two subscales: curiosity, which reflects an attitude of interest (e.g., "I am curious about each of my thoughts and feelings as they occur"), and decentring, which emphasizes maintaining a certain distance and identity, preventing oneself from being overwhelmed by one's thoughts and feelings (e.g., "I am receptive to observing unpleasant thoughts and feelings without interfering with them"). Higher scores on each subscale indicate higher levels of the respective construct. The total score represents the participant's level of mindfulness, with higher scores indicating higher levels of rmindfulness (Lau et al., 2006).

Igroup Presence Questionnaire (IPQ)

The Igroup Presence Questionnaire (IPQ) was used to measure participants' sense of presence during the virtual experience. The questionnaire includes 13 items and employs a seven-point rating scale (1 = Strongly disagree to 7 = Strongly agree). The questionnaire consists of three subscales (Schubert et al., 2001): spatial presence, which reflects the sensation of being physically present in the virtual environment (e.g., "In the virtual environment I had a sense of being there"); involvement, which measures the degree of engagement and immersion (e.g., "I was completely captivated by the virtual world"); and realism, which reflects the extent to which the virtual environment felt authentic (e.g., "How real did the virtual world seem to you?"). The total score represents the participant's overall sense of presence, with higher scores indicating a greater sense of immersion in the virtual experience.

Polar-H10

The Polar H10 chest strap captured heart rate (beats per minute) as well as interbeat intervals (milliseconds). Cardiac coherence was calculated following the method outlined by McCraty et al. (2009). The Kubios HRV software (Tarvainen et al., 2014) was used to calculate RMSSD and high-frequency HRV (HF-HRV). HRV quantifies the variation in consecutive heartbeats and serves as an indicator of SNS and PNS activity at any given time. HRV parameters are typically quantified through time-domain and frequency-domain measures. The primary time-domain measure is RMSSD, which reflects the variability in heart rate (HR) beats. It is commonly used to estimate vagally mediated changes in HRV. RMSSD is an established indicator of cardiac vagal tone (Berntson et al., 1997; 'Heart Rate Variability', 1996; Laborde et al., 2017). The primary frequency-domain measure is the high-frequency HRV (HF-HRV) component (0.15 to 0.40 Hz), which estimates vagally induced PNS input and the LF/HF ratio. In accordance with these standardized procedures, this study reports RMSSD and HF-HRV.

Results

Table 1 display descriptive statistics with means and standard deviations for the three groups. We applied a natural logarithm (ln) transformation to HF-HRV to overcome non-normality, enabling the use of parametric inferential statistical tests. A one-way ANOVA was conducted to investigate possible differences between the group's descriptive statistics. There was no significant age difference between the groups (F (2, 72) = 1.686, p = .192). For the questionnaire data, at pre-treatment there were no significant differences between the three groups (SAI: F (2, 72) = 1.820, p = .169); TMS: F (2, 72) = 1.369, p = .261); RMSSD: F (2, 72) = 1.654, p = .198); HF-HRV: F (2, 72) = 0.57, p = .467). We used repeated measures ANOVA to examine the effects of Group (VR-MG vs. MG vs. CG; between-subjects) and

Table 1 Descriptive statistics: between-group analyses for pre-treatment and post-treatment

	Virtual reality mindfulness mean (SE)	Mindfulness mean (SE)	Control mean (SE)
N (females)	25 (16)	25 (18)	25 (13)
Age (years)	19.52 (1.92)	19.76 (2.35)	20.68 (2.73)
SAI			
Baseline	63.00 (10.00)	61.88 (12.57)	57.52 (10.30)
Post-treatment	32.56 (6.67)	38.48 (12.57)	47.92 (11.30)
TMS			
Baseline	26.16 (8.12)	24.24 (7.00)	27.64 (6.66)
Post-treatment	33.92 (6.12)	29.52 (8.86)	29.92 (7.62)
RMSSD			
Baseline	30.26 (12.59)	39.38 (22.76)	34.81 (16.32)
Post-treatment	36.59 (14.50)	38.13 (18.26)	35.96 (15.91)
HF-HRV			
Baseline	5.888 (0.832)	6.182 (0.855)	6.094 (0.895)
Post-treatment	6.310 (0.679)	6.296 (0.820)	6.255 (0.814)

Time (pre vs. post; within-subjects). These analyses were conducted in IBM® SPSS® Statistics, Version 27.

STAI-S Scores

The state version of the State-Trait Anxiety Inventory scores (VR-MG, MG, and CG) revealed a significant main impact of Time in repeated measures ANOVA, (F (1, 72) = 145.356, p<.001, η^2 = 0.669, and a significant interaction between Time and Group, F (2, 72) = 12.302, p<.001, η^2 = 0.255. Further analysis also revealed simple effects of time, three groups were all significant (VR-MG: x̄= 30.64, SE = 3.048, p<.001; MG: x̄ = 23.4, SE = 3.048, p<.001; CG: x̄ = 9.6, SE = 3.048, p = .002) (Fig. 3).

We also computed the difference in degree of anxiety change for the three groups. A one-way ANOVA reveal significant differences among three groups (F (2, 72) = 12.302, p<.001). Bonferroni post hoc tests revealed that the degree of anxiety change in the VR-MG and MG was significantly greater compared to the CG group (x̄ = 21.04, SE = 4.31, p<.001; x̄ = 13.8, SE = 4.31, p = .006) (Fig. 3). There were no differences between VR-MG and MG (x̄ = 7.24, SE = 4.31 p = .292). In conclusion, all participants showed a decrease in anxiety after the manipulation. But virtual reality mindfulness and audio mindfulness training decreased significantly more scores for anxiety relative to listening to natural science articles.

TMS Scores

The Toronto Mindfulness Scale scores (VR-MG, MG, and CG) also revealed a significant main impact of time in repeated measures ANOVA, F (1, 72) = 44.639, p<.001, η^2 = 0.383, and a significant interaction between Time and Group, F (2, 72) = 4.297, p = .017, η^2 = 0.107. Further analysis also revealed simple effects of Time, with significant baseline and post-test results for the VR-MG and MG(VR-MG: x̄ = -7.76, SE = 1.324, p<.001; MG: x̄ = -5.28, SE = 1.324, p<.001), while CG's pre-and post-test results were not significant (CG: x̄ = -2.28, SE = 1.324, p = .089) (Fig. 3).

We calculated the difference in degree of mindfulness change across the three groups. A one-way ANOVA reveal significant differences among three groups (F (2,

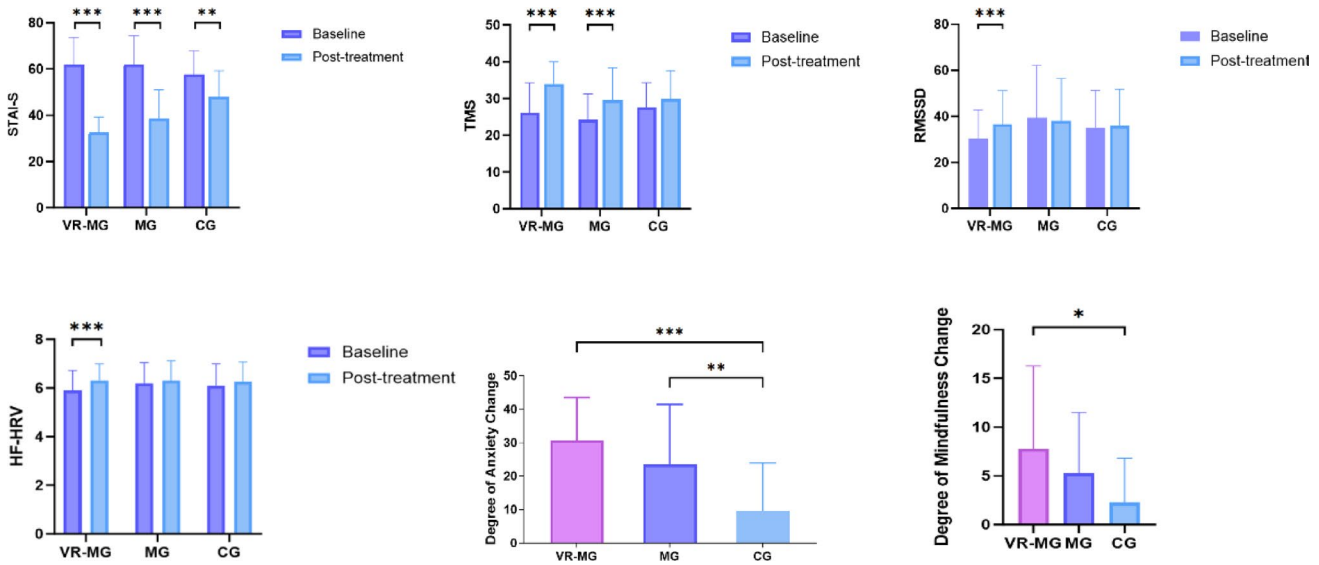

Fig. 3 Grouped histograms illustrating the changes in pre- and post-test scores for each indicator, as well as the differences in the degrees of anxiety and mindfulness changes across the three groups, were presented. The degree of anxiety change is calculated by subtracting the post-test anxiety score from the pre-test anxiety score. The degree of mindfulness change is calculated by subtracting the pre-test mindfulness score from the post-test mindfulness score. Note. Error bars are 95% CI. ***: P<.001; **: P<.01; *: P<.05..

72) = 4.297, p = .017). Bonferroni post hoc tests revealed that only the VR-MG and CG showed a significant difference in mindfulness state change ($\bar{x}$ = 5.48, SE = 1.87 p = .014) (Fig. 3). There were no significant differences between the VR-MG and MG groups ($\bar{x}$ = 2.48, SE = 1.87 p = .568), as well as between the MG and CG groups ($\bar{x}$ = 3.00, SE = 1.87, p = .34). In conclusion, virtual reality mindfulness and audio mindfulness training can effectively improve mindfulness state. But, only the change in mindfulness state in the VR-MG was significantly greater than that in the CG.

HRV

RMSSD

We conducted a repeated ANOVA with the factors Time (pre vs. post; within-subjects) and Group (VR-MG vs. MG vs. CG; between-subjects) to investigate the effect of the manipulation on participants' heart rate variability (see Fig. 3). The analysis revealed a significant main impact of Time, F (1, 72) = 4.774, p = .032, η^2 = 0.062, and a significant interaction effect of Group × Time, F (2, 72) = 5.561, p = .006, η^2 = 0.134. Sidak adjusted pairwise comparisons revealed that the RMSSD values in the showed a significant change between the pre-test and post-test (VR-MG: $\bar{x}$ = -6.334, SE = 1.645, p < .001), but the pre- and post-test results for the MG and CG were not significant. (MG: $\bar{x}$ = 1.257, SE = 1.645, p = .447; CG: $\bar{x}$ = -1.149, SE = 1.645, p = .487). Hence, Only the VR-MG group showed a significant increase in RMSSD over the course of the VR mindfulness training.

HF-HRV

We conducted a repeated ANOVA with the factors Time (pre vs. post; within-subjects) and Group (VR-MG vs. MG vs. CG; between-subjects) to investigate the effect of the manipulation on participants' HF-HRV (see Fig. 3). The analysis revealed a significant main impact of Time, F (1, 72) = 14.750, p = < 0.001, η^2 = 2.023, but no significant interaction effect of Group × Time, F (2, 72) = 2.500, p = .089, η^2 = 0.343. Further Sidak adjusted pairwise comparisons revealed that the HF-HRV values in the VR-MG showed a significant change between the pre-test and post-test (VR-MG: $\bar{x}$ = -0.421, SE = 0.105, p = < 0.001), but the pre- and post-test results for the MG and CG were not significant. (MG: $\bar{x}$ = -0.114, SE = 0.105, p = .281; CG: $\bar{x}$ = -0.151, SE = 0.105, p = .128. Hence, Only the VR-MG group showed a significant increase in HF-HRV over the course of the VR mindfulness training.

Table 2 Correlations among the changes in mindfulness, anxiety and HRV

	TMS change	SAI change	RMSSD change	HF-HRV change
TMS change	1.000			
SAI change	−0.432*	1.000		
RMSSD change	0.103	−0.012	1.000	
HF-HRV change	0.402*	−0.129	0.611**	1.000

***p < .001, **p < .01, *p < .05

Correlations Among the Changes in Mindfulness, Anxiety and HRV

Pearson correlation was used to analyze the relationships among mindfulness, anxiety and HRV changes in the VR-MG. The results showed that the changes in HF-HRV were positively correlated with the changes in TMS total score (r = .402, p = .047). The positive correlations indicated that greater increases in TMS were associated with greater reductions in negative emotions. No significant correlations were found in both the MG and CG (p > .05) (Table 2).

Mediation Effects

Further, we tested the mediation effects of the changes in RMSSD and HF-HRV between the changes in mindfulness levels and changes in anxiety levels following the VR mindfulness training. The results showed that the changes in RMSSD and HF-HRV did not mediate the relationship between the changes in mindfulness levels and changes in anxiety levels (Fig. 4). However, the direct effect of the change in mindfulness levels on the change in anxiety levels was significant ((a):β = -0.656, 95% CI: -1.260 to -0.052;(b):β = -0.683, 95% CI: -1.339 to -0.027).

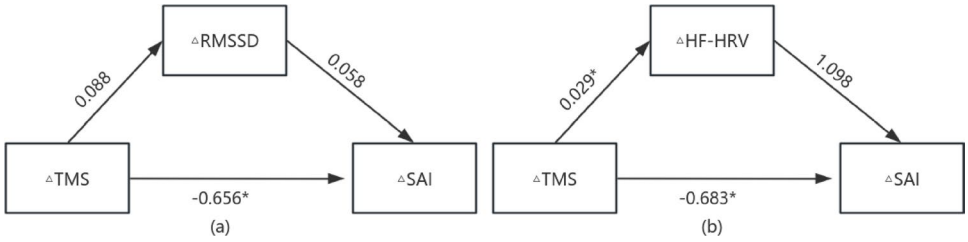

Fig. 4 A mediation model diagram showing the relationships between HRV changes, mindfulness changes, and anxiety changes

Participants' Experiences

We used the IPQ scale (Igroup Presence Questionnaire) to measure the immersion of the virtual experience. The average scores on Spatial Presence, Involvement, and Realism were 4.48 (SD = 4.27), 1.20 (SD = 4.14), and −1.24 (SD = 4.29), respectively. The results indicate that the game achieves the average level of spatial presence and involvement but performs poorly in terms of realism. This result is reasonable considering some participants reported that the scene's visuals were more aesthetically pleasing with a casual cartoon style.

When asked about their experience after the intervention in the VR-MG, 80% (20/25) of participants reported positive experiences with virtual reality mindfulness, describing feelings of relaxation, calmness, and novelty. For example, participant #12 stated, "I felt very relaxed because I rarely experience nature, and the scene just now made me feel like I was taken to a place beyond reality." In contrast, 8% (2/25) had negative experiences, citing factors such as the VR device being too heavy, not liking the tone of the guided instructions, and the scenes not being realistic enough. For instance, participant #15 mentioned, "I think the scene was too cartoonish. I would be more willing to use it if it were more realistic."

Discussion

This study explores the interplay between mindfulness, anxiety, and heart rate variability. It involved a sample of university students experiencing recent anxiety, who were randomly assigned to either virtual reality mindfulness training, mindfulness training, or an active listening control condition. Through the design of a study in which participants engaged in a five-minute interactive VR mindfulness training, we tested the effects of mindfulness on anxiety. The study tested three hypotheses: H1) Compared to pre-training, the VR-MG would experience a reduction in anxiety and an increase in mindfulness levels and HRV following training; H2) The VR-MG would show more effective training outcomes compared to traditional audio-based mindfulness training; H3) HRV mediates the improvement in anxiety induced by mindfulness.

The study results indicate that after a single five-minute intervention, the interactive virtual reality-based mindfulness intervention significantly reduced anxiety levels, improved mindfulness, and enhanced HRV response. As predicted, partial support for H2 was found. Compared to the traditional mindfulness audio group, the VR-MG showed a more significant improvement in mindfulness levels than the active control group. In terms of HRV, significant pre- and post-intervention effects were observed only in the VR-MG. However, regarding anxiety reduction, although the VR-MG showed a greater decrease in anxiety compared to the MG, no significant difference was found between the two groups in terms of improvement. Additionally, no support for H3 was found, as significant changes in HRV were observed only in the VR-MG during the intervention. We then explored potential mediating effects within the VR-MG, and the results indicated that mindfulness intervention had a significant direct effect on anxiety reduction, but HRV did not mediate the influence of mindfulness on anxiety.

The findings align with prior research demonstrating the efficacy of brief mindfulness interventions in alleviating anxiety (Goldberg et al., 2018; Spijkerman et al., 2016). Notably, the unique enhancement of HRV in the VR-MG suggests that immersive virtual environments may amplify physiological regulation by engaging multisensory inputs, thereby fostering deeper attentional focus (Navarro-Haro et al., 2017; Riva et al., 2021). The interactive design of the VR environment, which incorporated real-time biofeedback (e.g., HRV-driven visual and auditory changes), likely reinforced participants' awareness of their physiological state, facilitating parasympathetic activation and vagal tone improvement (Shaffer et al., 2014). This aligns with studies showing that biofeedback-integrated VR systems enhance self-regulation by bridging internal states with external stimuli (Arpaia et al., 2022). The results indicate that the VR-MG outperformed the MG in enhancing mindfulness levels. This may stem from VR's capacity to reduce environmental distractions and enhance immersion, allowing participants to anchor attention more effectively to guided instructions (Cipresso et al., 2018). The visually engaging, nature-inspired VR environment, designed based on Attention Restoration Theory (Kaplan, 1995), likely amplified the restorative effects of mindfulness by evoking positive emotional responses and reducing cognitive load (Browning et al., 2020). Such immersive experiences may lower the cognitive effort required for beginners to sustain focus, thereby optimizing mindfulness acquisition (Nararro-Haro et al., 2016). However, the lack of a significant difference in anxiety reduction between VR-MG and MG parallels previous studies reporting comparable effects of virtual and traditional mindfulness modalities on emotional outcomes (Gentile & Kim, 2024; Zhang et al., 2023). This implies that while VR enhances adherence and immersion (Modrego-Alarcón et al., 2021), its added value in anxiety reduction may depend on individual preferences or baseline anxiety severity.

The differential effects on HRV parameters warrant interpretation. While RMSSD showed a significant Group × Time interaction, HF-HRV demonstrated only a main effect

of time. This divergence aligns with evidence that time-domain (RMSSD) and frequency-domain (HF-HRV) measures capture distinct autonomic responses (Berntson et al., 1997). RMSSD primarily reflects rapid vagal modulation, making it sensitive to immediate parasympathetic activation during biofeedback-enhanced mindfulness (Kirk & Axelsen, 2020). In contrast, HF-HRV's strong coupling to respiratory patterns (Grossman & Taylor, 2007) may explain its limited group differences: the VR environment's guided breathing cues (4-second particle rhythm) likely standardized respiration across participants, reducing between-group variability (Tarvainen et al., 2014). Methodological considerations, including spectral analysis constraints, further support prioritizing RMSSD as a marker for brief VR interventions.

The absence of mediation by HRV challenges the assumption that autonomic nervous system modulation is a primary pathway through which mindfulness reduces anxiety. Instead, anxiety reduction may stem from cognitive mechanisms, such as reduced rumination or improved emotional regulation (Hofmann & Gómez, 2017), which were not directly measured here. This aligns with neurocognitive models positing that mindfulness enhances prefrontal cortical control over limbic reactivity (Davidson & Kaszniak, 2015), independent of HRV-mediated pathways.

Several limitations warrant consideration. First, the single-session design precludes conclusions about long-term efficacy. Longitudinal studies are needed to determine whether repeated VR mindfulness sessions yield sustained improvements in HRV and anxiety. Second, the homogeneous sample of university students limits generalizability to clinical populations or older adults with chronic anxiety. Future research should explore these interventions in individuals with diagnosed anxiety disorders or age-related autonomic dysregulation. Third, the reliance on self-reported anxiety measures introduces potential response biases. Incorporating objective biomarkers, such as cortisol levels or neural activity, could provide a more comprehensive understanding of mindfulness mechanisms. Furthermore, the cartoonish aesthetic of the VR environment, criticized by some participants, may have limited realism and presence. Future designs could prioritize photorealistic environments to enhance immersion and emotional resonance. Despite these limitations, the study offers practical insights. The rapid efficacy of a five-minute VR intervention highlights its potential as a scalable tool for acute anxiety management in educational or workplace settings. Furthermore, the differential effects on HRV suggest that VR mindfulness may be particularly beneficial for individuals with autonomic dysfunction, such as those with stress-related cardiovascular conditions.

Conclusion

In summary, this study demonstrates that a single five-minute virtual reality (VR)-based mindfulness intervention significantly reduces anxiety, enhances mindfulness levels, and improves heart rate variability (HRV) in university students. Notably, the VR mindfulness group (VR-MG) exhibited unique physiological benefits, with significant pre-post improvements in both RMSSD and HF-HRV, which were absent in the traditional mindfulness (MG) and control groups (CG). These findings suggest that immersive VR environments, integrated with real-time biofeedback, may amplify parasympathetic activation and autonomic regulation, offering a dual pathway for psychological and physiological restoration. While VR-MG and MG showed comparable reductions in anxiety, VR-MG demonstrated a marginally greater improvement in mindfulness levels compared to MG, though this difference did not reach statistical significance. Contrary to expectations, HRV did not mediate the relationship between mindfulness and anxiety reduction. In brief, the immediate efficacy of a five-minute VR intervention underscores its potential as a scalable strategy for acute stress mitigation in academic and occupational environments. It is our aspiration that educators and clinicians could integrate brief VR mindfulness sessions into daily routines to mitigate acute anxiety while minimizing time and resource burdens.

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Author Contributions Xu and Gu made substantial contributions to the conception and design of the work. Xu wrote the main manuscript, prepared all figures, and conducted the experiments. Gu supervised the entire project. Hu developed the new software used in the work. All authors reviewed and approved the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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